



CanSat Competition 2026 Rulebook - Judging & Evaluation

1. General Rules

- Teams must consist of **3-5 students**.
- The CanSat must be **self-built during the build phase**. Use of pre-fabricated kits is not permitted.
- The CanSat must comply with the following physical limits:
 - Maximum dimensions: **21 cm (+7cm [max], for egg chamber) × 12 cm**
 - Maximum mass: **500 g (±10%)**
- Each CanSat must include the following mandatory systems:
 - **Egg payload** (fragile payload integrity test).
 - **Descent system** (parachute or equivalent).
 - **Telemetry system** with altitude, temperature, gyroscope, and accelerometer sensors.
- **Mandatory Ground Station Compatibility**
All participating CanSats must be compatible with at least one of the Ground Stations provided by the Physics Club and must be capable of successfully transmitting the required telemetry data during the mission.
- The Physics Club provides two Ground Station communication options:
 1. 433 MHz LoRa Ground Station - using the SX1278 LoRa RA-02 module.
 2. nRF24L01 Ground Station - using the nRF24L01 module.
- Teams are responsible for ensuring that their CanSat's communication system is compatible with one of the above Ground Stations and that reliable data transmission is achieved.

2. Mission Profile

The competition mission consists of the following steps:

- The CanSat will be launched to an altitude of **100 ft** (equivalent to an 8-story building).
- **Parachute deployment** must be clearly demonstrated.
- The CanSat must descend safely and protect the egg payload from breakage.
- Telemetry data must be transmitted continuously to the ground station during flight.

2A. Launch Guidelines

- The launch will take place from the **Drone** at a height of **100ft**.

2B. Event Flow of Launch

- Each CanSat must be **powered on at the ground floor** before launch. At this stage, the telemetry system should report an altitude of **0 (or near 0)** to confirm baseline calibration.
- The CanSat will then be **carried through the Drone in at certain heights**. Telemetry must reflect this altitude change.
- From the Drone, the CanSat will be **released for descent**, with parachute deployment and stable telemetry transmission expected.
- After landing, payload recovery and final data logging will be verified.

3. Evaluation Criteria (200 Points Total)

A. Payload Safety - 25 Points

- **Egg Payload Integrity (20 Points Total):**
 - Egg payload recovered **unbroken** → 20 points.
 - Egg payload with **minor cracks** → 15 points.
 - Egg payload with **major cracks** → 5 points.
 - Egg payload **completely broken** → 0 points.
- **Parachute Deployment (5 Points):**
 - Proper and timely parachute deployment during descent → 5 points.

B. Telemetry & Communication - 25 Points

Real-Time Data Transmission (20 Points):

- Continuous transmission is mandatory from the moment the CanSat is **powered on at the ground floor**, during **lift-up to launch altitude**, throughout **flight**, and after **landing until recovery**.
- The CanSat must transmit at least **1 packet per second** during the entire operation.

Transmission Capability & Reliability (5 Points):

- Higher packet rates will be rewarded with more points, provided transmissions remain consistent.
- Packet loss will reduce the score — teams must balance **high transmission frequency with minimal loss**.
- All performance will be measured using the **Physics Club's official dual ground stations**.

Data Format Compliance (Mandatory):

- All transmitted data must strictly follow the **specified packet format**.
- The **correct team number (CAN-Team-XX)** must be included in every packet.
- If the format or team number is incorrect, **no points will be awarded for telemetry**, regardless of transmission success.

Valid Packet Definition:

- A valid packet is one that contains at least the **team number** and **packet number**, along with the following **minimum required parameters**:

C. Parachute, Descent, Stability & Integrity – 25 Points

1. Descent Rate & Stability (6 Points):

- CanSat must descend at a rate of ≤ 5 m/s for maximum safety.
- Stable descent without **tumbling or uncontrolled spinning** will be rewarded.
- **Time of Flight Consideration:**
Teams with **longer, stable descent times** (without exceeding safety conditions) will be scored relatively higher.
This factor will be evaluated comparatively across teams to encourage **controlled and safe descent profiles**.

2. Sensor Data Continuity (9 Points):

- Telemetry data from onboard sensors (altitude, temperature, pressure, orientation, and accelerometer) must remain **consistent and accurate** throughout the descent.
- Missing or corrupted data streams will reduce points.

3. Structural Integrity & Post-Landing Transmission (5 Points):

- The CanSat's **structure must not be damaged** upon landing (minor scratches allowed).
- One can use any safety system according to their innovation to soft land their CanSat.
- The system must continue transmitting data for at least **5 seconds after impact** to ensure survivability.

4. Bare Minimum Functional Requirements (5 Points):

- Each CanSat must include:
 - A **manual switch** to turn the system on/off.
 - A **visible LED indicator** confirming power-on status.
- Missing either of these will result in loss of points.

D. Structural & Material Innovation – 30 Points

1. Compactness & Efficiency (5 Points):

- Effective utilization of size and mass constraints (≤ 21 cm $\times$ 12 cm, ≤ 500 g).
- Optimized internal layout for components, payload protection, and wiring.
- Efficient use of volume to maximize performance in limited space.

2. Structural & Material Design (5 Points):

- Clear documentation of structural design with CAD models/diagrams.
- Justification of material choices (lightweight, durable, cost-effective).
- One can use any material for outer structure of CanSat (PVC, Plastic, 3D print)
- Bonus for sustainable or unconventional materials (3D printed, recycled, composites).

3. Fabrication & Innovation (5 Points):

- Creative construction, modularity, or detachable systems.
- Evidence of hands-on fabrication (not just off-the-shelf assembly).
- Innovative solutions for housing, recovery, or integration.

4. Aesthetics & Build Quality – 15 Points

- Overall appearance, neatness, and professional finish of the CanSat.
- Proper cable management and minimal clutter.
- Attention to detail in assembly, labeling, and external presentation.
- Bonus for a visually appealing yet functional design.

⇒ Total = 30 Points

E. Technical Design & Analysis – 70 Points

1. PCB Design – 15 Points

- Original, well-documented PCB design with clear layout diagrams and views.
- Minimal external wiring; well-routed traces and compact integration.
- High soldering quality and professional assembly.
- Bonus for fully custom boards made for CanSat (instead of generic dev boards).

2. Code Originality – 10 Points

- Self-written, well-commented, and functional code.
- Efficiency and reliability in mission-critical functions.
- Reuse of open-source libraries acceptable only with modifications and proper documentation.

3. Sensor Integration – max 25 Points

- **Bare Minimum Sensors (mandatory, 15 points):** MPU6050 (gyroscope + accelerometer), BMP (pressure + altitude), LoRa (telemetry).
- **Additional Working Sensors** (5 Points each, up to 25 Points). Examples: magnetometer, GPS, humidity, air quality, UV, Camera module, custom/innovative sensors.

- Non-working but properly integrated and documented sensors may receive partial credit (2 points).

4. Data Analysis – max 20 Points

- Teams will have **4 hours post-launch** to analyze ground station data.
- Mandatory graphs: **altitude, temperature, and pressure** vs time / packet no.
- Extra credit for: acceleration profiles, orientation changes, descent rate, correlations, or other derived insights.
- Creativity, clarity of presentation, and depth of interpretation will be rewarded.
- Clear visualization methods (plots, comparative graphs, derived metrics).
- Strong interpretation of results tied to CanSat performance.
- Professional presentation and unique analytical insights.

F. Final Report – 25 Points

The final report is the complete record of the CanSat project. It must include design details, mission execution, analysis, documentation, and supporting media. Reports will be judged on completeness, clarity, and professionalism.

1. Structure & Documentation – 20 Points

- Report must include: design approach, system architecture, mission procedure, results, and lessons learned.
- All supporting materials (schematics, PCB files, CAD designs, wiring diagrams) must be attached or referenced.
- Proper citations for any external resources used.
- All graphs generated during data analysis must be compiled into the report.

2. Imaging & Media Documentation – 5 Points

- High-quality images of CanSat (top, side, bottom views).
- PCB documentation with multiple views (top, bottom, side).
- Team photo with the CanSat and group photo with mentors along with CanSat are mandatory.
- All required photos and media must be presented in the report.

➡ **Total = 25 Points**

4. Final Data Analysis

- After launch, ground station data will be shared with each team.
- Teams will have **4 hours** to analyze and prepare their findings.
- Submissions will be evaluated on **accuracy, completeness, and creativity** in presenting results.

5. Disqualification Conditions

A team will be disqualified if:

- Size or weight exceeds the specified limit by more than 10%.
- Deployment is unsafe (projectile motion, no parachute, uncontrolled crash).
- No attempt at a communication system is made.
- The team arrives late at the venue.
- Code of conduct is violated.

6. Winner Determination

- The winner is the team with the **highest cumulative score out of 200**.
- The runner-up is the team with the **second-highest score**.
- In case of a tie, the following priorities will apply:
 1. Payload Safety
 2. Telemetry Accuracy
 3. Descent Stability

7. Data Transmission Guidelines

All teams must follow a **standardized data packet format** for telemetry.

Packet Format:

- CAN-Team-XX; P-XXX; Ti-HH:MM:SS:MS; A-XXX.X; Pr-XXXX.XX; T-XX.X; Ro-XX.X; Pi-XX.X; Ya-XX.X; AX-XX.XX; AY-XX.XX; AZ-XX.XX;

Field Descriptions:

Field	Description	Units	Precision
CAN-Team-XX	Unique team identifier (e.g., CAN-Team-01)	–	–
P-XXX	Packet count/number (increments per packet)	–	Integer
Ti-HH:MM:SS:MS	Timestamp (Hour:Minute:Second:Millisecond)	–	–
A	Altitude	m	1 decimal
Pr	Pressure	Pa	2 decimals
T	Temperature	°C	1 decimal
Ro	Roll	°	1 decimal
Pi	Pitch	°	1 decimal
Ya	Yaw	°	1 decimal
AX	Acceleration X	m/s ²	2 decimals
AY	Acceleration Y	m/s ²	2 decimals
AZ	Acceleration Z	m/s ²	2 decimals

Optional Sensor Fields

Teams may add extra sensors, but must use short prefixes to save bandwidth. Priority must always be given to transmitting mandatory fields first.

Examples of additional sensor prefixes:

- GP-Lat, GP-Lon, GP-Alt → GPS Latitude, Longitude, Altitude
- MQ → MQ-135 Air Quality (ppm)
- MGX, MGY, MGZ → Magnetometer X, Y, Z (μ T)
- ESP32 CAM WiFi Module → Live Telemetry

Sample Packet with Additional Sensors

CAN-Team-01; P-025; Ti-14:32:45:120; A-102.3; Pr-100534.25; T-24.7;
Ro-1.5; Pi-0.8; Ya-2.3; AX-0.02; AY-0.15; AZ-9.81; GP-Lat-21.1702;
GP-Lon-72.8311; MQ-135-180.25; MGX-35.2; MGY-12.8; MGZ-5.6;

Notes for Teams

1. **Mandatory data must always be transmitted first.** Optional fields may be appended if bandwidth allows.
2. Prefixes should be **short, standardized, and minimal** to reduce payload size.
3. Packet numbers must start from **P-001** and increment sequentially.
4. Missing or corrupted mandatory fields → **no points** for telemetry.
5. Extra sensors will be rewarded under the **Sensors & Innovation scoring section**, but only if data is correctly formatted and consistently transmitted.

8. Minimum CanSat & Transmission Requirements

To ensure fair communication and avoid interference, the following rules apply:

- **Power Indicator:** Each CanSat must turn on an **LED indicator** immediately upon power-on to confirm activation.
- **Automatic Data Transmission:** Telemetry transmission must begin **as soon as the CanSat is powered on** without manual triggering.
- **Radio Protocol & Sync Word:**
 - During official launch, all teams must configure LoRa modules with **sync word = 0xA5**.
 - For testing before launch, teams must change the sync word to **0xF3**.
 - Any team transmitting with the wrong sync word during another team's launch will be penalized.
- **Interference Policy:**
 - While one team's CanSat is being launched, **all other teams must keep their CanSats turned off**.
 - If packets from another team are received during a launch, a penalty of **−1 point per 2 packets** will be applied.
- **Transmission Frequency:** A minimum rate of **1 packet per second** is required. Higher rates are allowed if stable and interference-free.
- **Ground Stations:** The Physics Club will provide and operate **two official ground stations** to ensure accurate reception and recording of all telemetry packets.

Egg Chamber Requirement:

Each CanSat must include a **cushioned and secure chamber** specifically designed to hold the egg payload. The chamber must protect the egg during launch, descent, and landing. The chamber must also stay within the updated CanSat dimension limits of **21 cm (+7 cm [max] for egg chamber) × 12 cm**.

Parachute Placement:

The parachute must be positioned in such a way that it **deploys instantly upon launch/release**. It must not be tightly packed inside the structure, but instead integrated externally or semi-exposed for immediate deployment.

9. Report & Video Submission Process

Report Submission Process:

- A Google Form link will be provided for all report submissions.
- Teams must upload their CanSat Preliminary Report (excluding data analysis) and Final Report (with analysis) through this form.
- The report file must be uploaded as a **Google Docs link**, with file size **not exceeding 30 MB**.
- Ensure access permissions are set to *Anyone with the link can view*.

Creative Video Submission Process:

- Each team must prepare a short video documenting their CanSat project.
- The video must be uploaded to:
 1. **Team member's YouTube channel**
 2. **Team member's Instagram page** (as a post/reel)
- Both video links (YouTube & Instagram) must be pasted into the **Google Form** for video submission.
- **Tag the Physics Club, SVNIT** on both YouTube and Instagram posts to ensure official acknowledgment.

Prize Money:

- The announced prize money will be distributed among the 1st, 2nd, and 3rd position teams as per applicable **SVNIT norms, rules, and tax deductions**. Terms & Conditions apply.
- The decision of the Judges and Physics Club shall be final and binding in all matters related to the competition.